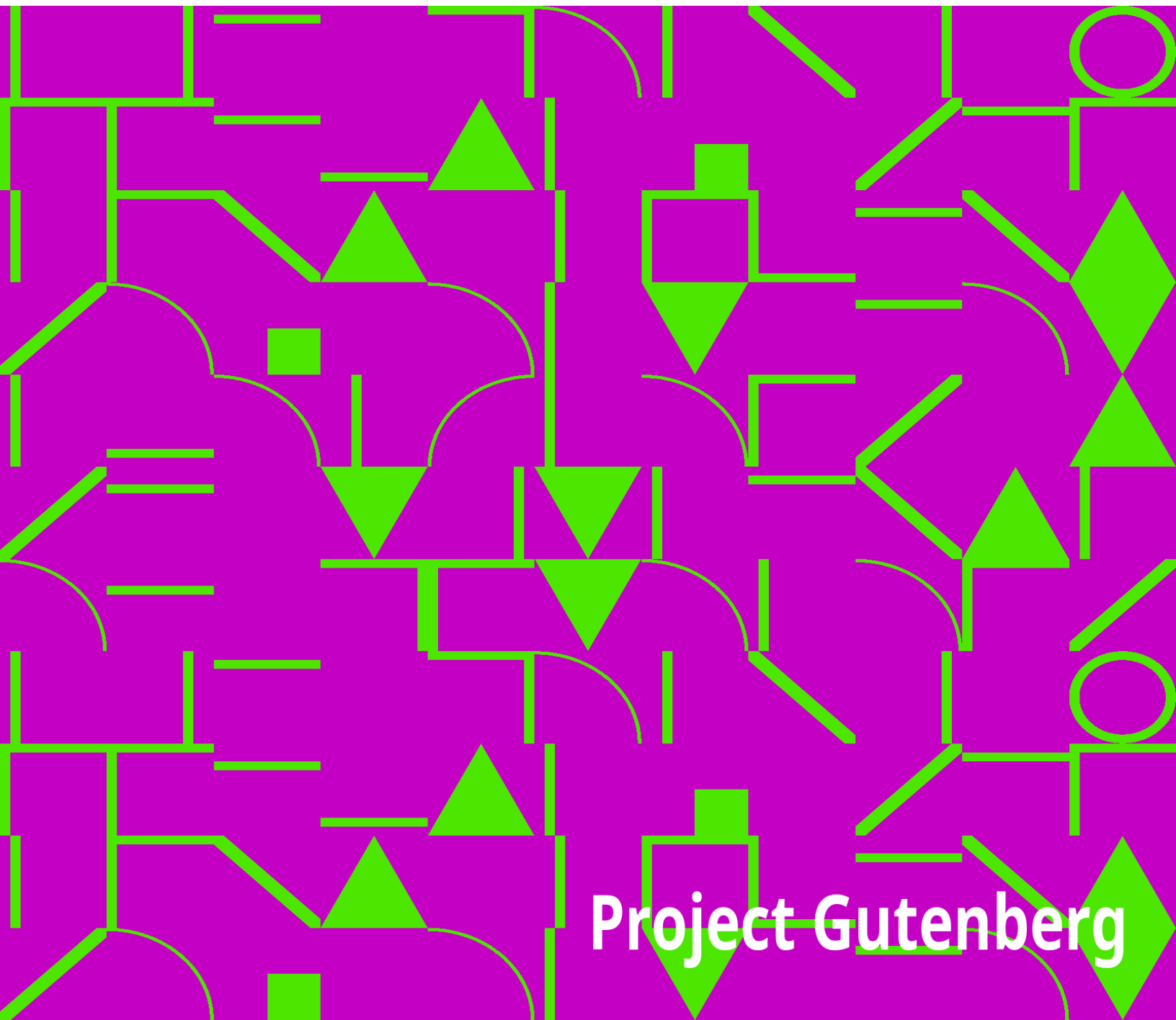


The Pigmy Woodrat, *Neotoma goldmani*, Its Distribution and Systematic Position

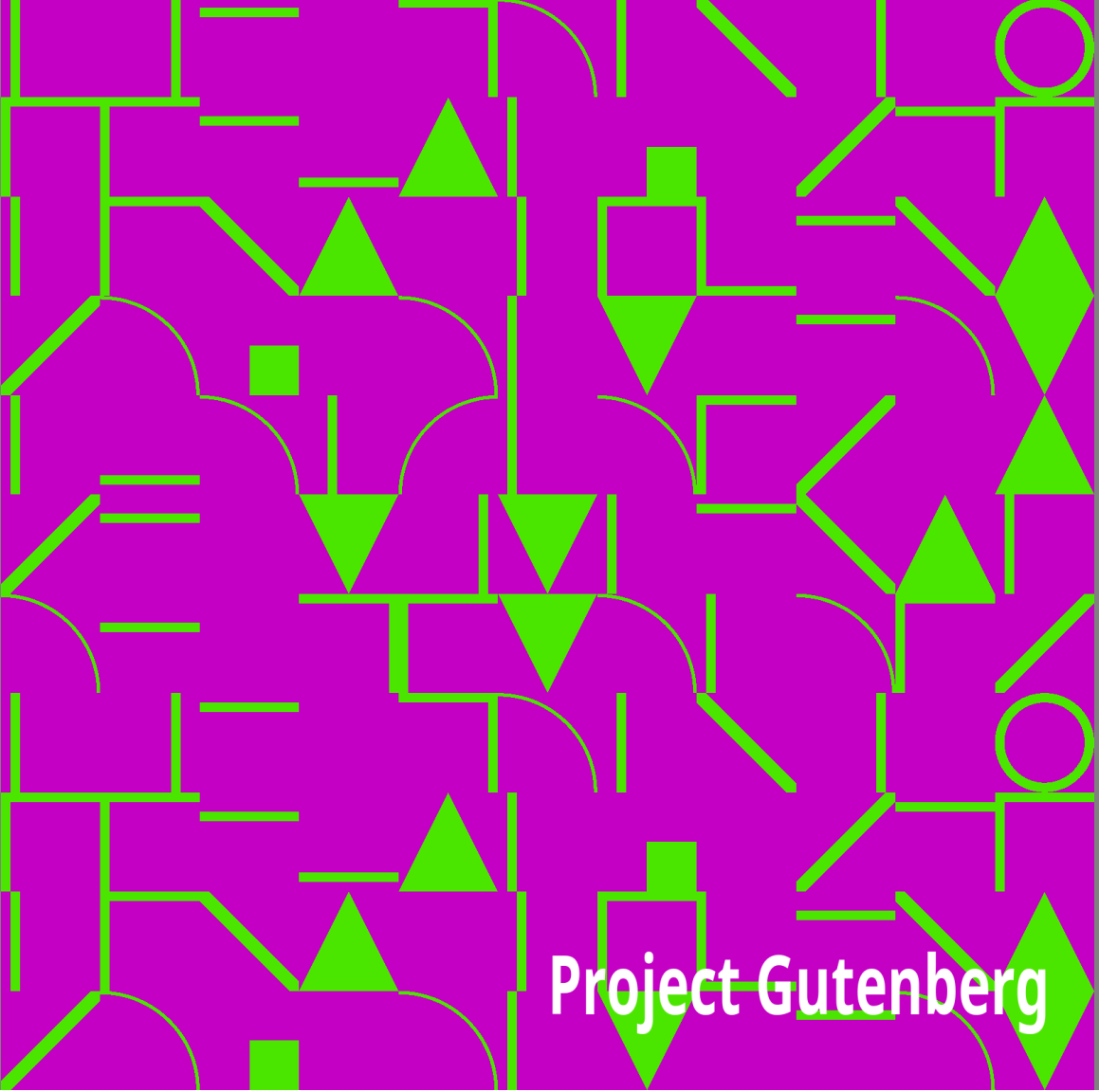
Dennis G. Rainey and Rollin H. Baker



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An abstract geometric pattern consisting of various shapes like triangles, squares, circles, and lines in shades of blue, green, and yellow, set against a dark background.

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BY
DENNIS G. RAINEY AND ROLLIN H. BAKER

UNIVERSITY OF KANSAS
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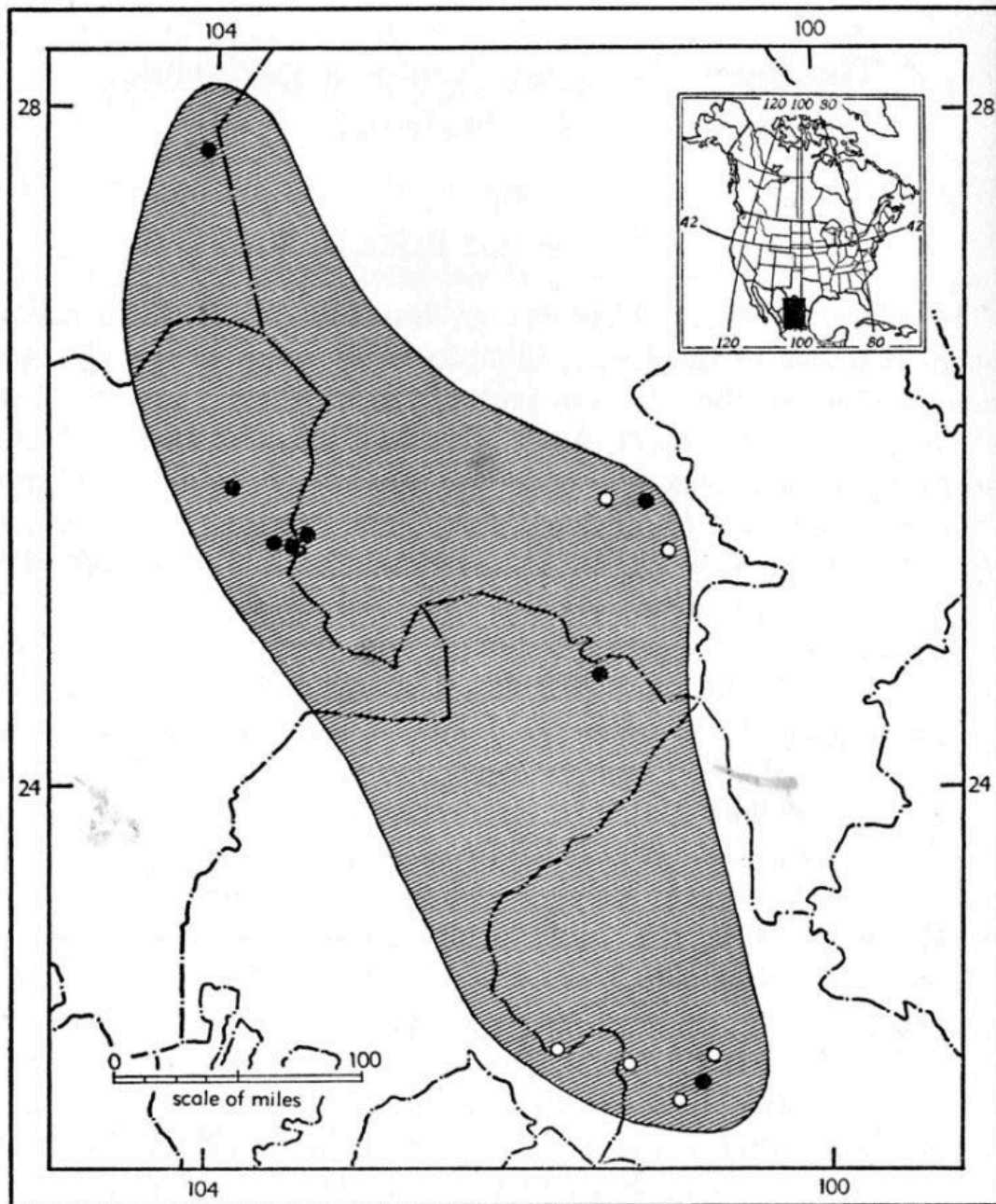
Dennis G. Rainey and Rollin H. Baker

The pigmy woodrat, *Neotoma goldmani* Merriam, the smallest known member of the genus, inhabits rocky areas in the elevated desert regions of the northern part of the Mexican Plateau (Mesa del Norte). Goldman (N. Amer. Fauna, 31:82, October 10, 1910) had for study ten specimens from two localities in Coahuila. Since his report, Dalquest (Louisiana State Univ. Studies, Biol. Sci. Ser. No. 1:162, December 28, 1953) extended the known distribution of this species approximately 225 miles southward into San Luis Potosí, where he reported animals from five localities. Field workers from the Museum of Natural History at the University of Kansas recently have taken *goldmani* in the Mexican states of Chihuahua, Coahuila, Durango, Zacatecas and San Luis Potosí, and now we can define, with greater accuracy, the geographic range of this species (see fig. 1 and list of specimens examined).

Goldman (*loc. cit.*), relying chiefly on external appearance, placed *goldmani* in the *desertorum* group, now known as the *lepida* group (Goldman, Jour. Mamm., 13:67, February 9, 1932). Blossom (Occ. Papers Mus. Zool., Univ. Michigan, 315:3, May 29, 1935) thought that *goldmani* might be a subspecies of *lepida* but that intergradation between the two had not been demonstrated. Our newly acquired material, instead of confirming the opinions of Goldman and Blossom, shows that *goldmani* is more closely related to *Neotoma albigula*.

Externally *goldmani* resembles *Neotoma lepida* (examples from California, Utah, and Colorado) in having long, silky pelage; ochraceous buffy coloring, especially along sides; and underparts basally plumbeous except for a small throat patch where the hairs are entirely white in some

individuals. In *albigula* this patch of white hairs usually is much larger and more conspicuous. Cranially, instead of resembling the *lepida* group (including *Neotoma stephensi*), *goldmani* looks more nearly like a miniature *albigula* (specimens of *albigula* from Coahuila). The auditory bullae, in relation to the length of the skull, are of comparable size in *goldmani* and *albigula* whereas those of the *lepida* group are proportionately much larger. Moreover, the posterior margin of the palatal bridge is concave in *goldmani* and *albigula* instead of truncate as in the *lepida* group. *Neotoma goldmani* differs from both *albigula* and *lepida* in: ascending branches of premaxillaries broader posteriorly; supraorbital ridges less pronounced; rostrum less massive; interparietal broader in relation to width of cranium; interorbital space, relative to length of skull, wider; and upper molar teeth broader in relation to their length.



**FIGURE 1. Distribution of the pigmy woodrat, *Neotoma goldmani*.
Solid circles represent specimens examined; hollow circles
represent others reported but not examined.**

The baculum of *goldmani*, when compared with bacula and with figures of these bones in Burt and Barkalow (Jour. Mamm., 23:291 and 293, August 13, 1942) of species representing the *floridana*, *lepida*, *albigula*, *mexicana*, *fuscipes*, and *cinerea* groups, was found to resemble most closely the baculum of *albigula* in general proportions (ratio of length to lateral

diameter of base) and in having a distinct knob at the distal end. The baculum of *goldmani* differs slightly from that of *albigula* in having a less downwardly curved shaft and in having a less pronounced median dorsal depression at the proximal end. Although *goldmani* bears some external resemblance to *lepida*, the cranial characters mentioned above and the size and shape of the baculum show that *goldmani* is best arranged as a member of the *albigula* group.

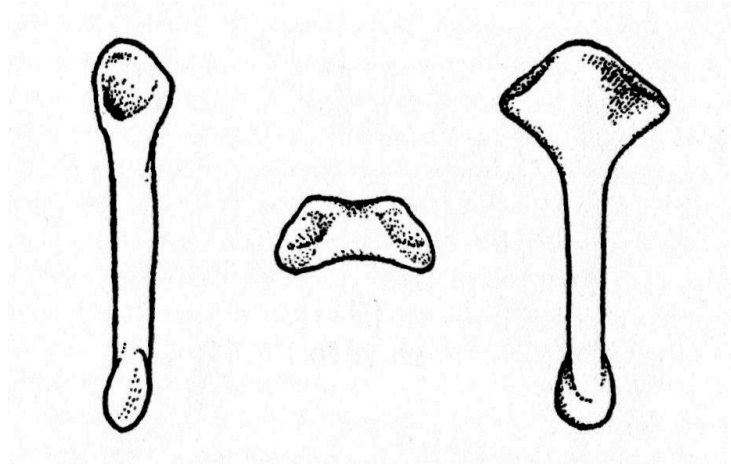


FIGURE 2. Dorsal, end (proximal), and lateral views of the baculum of *Neotoma goldmani*, adult, No. 40758 KU, $\times 5$.

Measurements (in millimeters) of the mature baculum (fig. 2, No. 40758 KU) are: total length, 6.2; lateral diameter of base, 2.6; dorso-ventral diameter of base, 1.4; lateral diameter of the shaft near the middle of the bone, 0.6. Except for being smaller, the bacula of the younger *goldmani* are like the mature ones.

Assistance with field work is acknowledged from the Kansas University Endowment Association and the National Science Foundation. The figures were drawn by Victor Hogg. Bacula were prepared for study following the method outlined by White (Jour. Mamm., 32:125, February 15, 1951).

Specimens examined.—Total, 15, all in the Museum of Natural History at the University of Kansas. Localities within any one state are arranged from north to south. *Chihuahua*: Sierra Almagre, 6000 ft., 12 mi. S Jaco, 1. *Coahuila*: 17 mi. N and 8 mi. W Saltillo, 5200 ft., 2; 3 mi. SE Torreón, 3800 ft., 7. *Durango*: 1 mi. SSE Mapimí, 4100 ft., 1; 4 mi. WSW Lerdo, 3800 ft., 1; 5 mi. SE Lerdo, 3800 ft., 1. *Zacatecas*: Conception del Oro, 7680 ft., 1. *San Luis Potosí*: 10 mi. NE San Luis Potosí, 6000 ft., 1.

Other records.—*Coahuila*: Jaral; Saltillo (Goldman, 1910:82). *San Luis Potosí*: Cerro Peñon Blanco; Ventura; Santa Teresa; city of San Luis Potosí (Dalquest, *loc. cit.*).

Transmitted February 8, 1955.

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